THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which of the following will have the highest H^+ ion concentration? 1

(a) pH 1 **(b)** pH 10 **(c)** pH 7 **(d)** pH 4

2. A metal M reacts with cold water giving hydrogen and a base. M could be: 1

(a) copper **(b)** sodium **(c)** gold **(d)** silver

3. The number of covalent bonds in propane C_3H_8 is: 1

(a) 10 **(b)** 8 **(c)** 11 **(d)** 9

4. Which of the following is NOT a requirement for photosynthesis? 1

(a) oxygen **(b)** sunlight **(c)** water **(d)** chlorophyll

5. The connection between sensory and motor neurons inside the spinal cord is made by: 1

(a) effector **(b)** dendrite **(c)** axon **(d)** relay neuron

6. A unicellular organism that divides into many cells inside a protective cyst is 1

(a) Plasmodium **(b)** Paramecium **(c)** Yeast **(d)** Amoeba

7. Two tall plants produce 75 tall and 25 dwarf plants. The parents were most likely 1

(a) both Tt **(b)** both TT **(c)** TT and Tt **(d)** Tt and tt

8. A concave mirror of f = 15 cm forms an image of the same size as the object. The object distance is 1

(a) 7.5 cm **(b)** 15 cm **(c)** 45 cm **(d)** 30 cm

9. A person's near point is 50 cm. To read at 25 cm, he needs a lens of power 1

(a) -2 D **(b)** +4 D **(c)** +2 D **(d)** +0.5 D

10. A charge of 600 C flows in 5 minutes. The current is 1

(a) 2 A **(b)** 0.5 A **(c)** 120 A **(d)** 3 A

11. The magnetic field produced by a straight wire at a distance 4 cm is B. At 2 cm from the wire (same current), the field is 1

(a) zero **(b)** smaller than B **(c)** larger than B **(d)** equal to B

12. Which of the following is the correct food chain? 1

(a) Plant, caterpillar, bird, hawk **(b)** Plant, hawk, bird

(c) Bird, plant, hawk **(d)** Hawk, bird, caterpillar, plant

13. In $ZnO + C \longrightarrow Zn + CO$, the reducing agent is: 1

(a) ZnO **(b)** CO **(c)** C **(d)** Zn

14. $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ is a: 1

(a) combination **(b)** thermal decomposition

(c) displacement **(d)** neutralisation

15. Electronic configuration of an element X is 2, 8, 2. It forms an ion of charge: 1

(a) +2 **(b)** +1 **(c)** -2 **(d)** 0

16. CH_3CH_2OH with alkaline $KMnO_4$ and heat gives: 1

(a) $CH_2{=}CH_2$ **(b)** CH_3COOH **(c)** CH_3CH_3 **(d)** $CH_3COOC_2H_5$

17. Anaerobic respiration releases energy in the: 1

(a) mitochondria **(b)** cytoplasm **(c)** chloroplast **(d)** ribosome

18. If the tip of a shoot is removed, phototropic bending: 1

(a) reverses **(b)** does not occur **(c)** occurs more **(d)** increases

19. A flower has only stamens. It is 1

(a) sterile **(b)** bisexual **(c)** unisexual (male) **(d)** female

20. A couple already has two daughters. The probability that their third child is a boy is 1

(a) 1/2 **(b)** 1/4 **(c)** 0 **(d)** 1

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A food chain has the organisms: grass, frog, grasshopper, snake. Arrange them in the correct order and identify the tertiary consumer. 2

22. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and name the gas evolved. 2

23. How does a farmer treat acidic soil? Give reason. 2

24. Why does an ionic compound conduct electricity in aqueous solution but not in the solid state? 2

25. How will you test for unsaturation in a hydrocarbon? 2

26. Why do herbivores have a longer small intestine than carnivores? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Compare the properties of soap and detergent. 3

28. Explain why the oxygenated and deoxygenated blood must not mix in the heart. 3

29. Describe the role of feedback in the control of thyroxine. 3

30. Why is there a wide difference in the number of male and female gametes in many organisms? 3

31. Calculate the number of seeds of each phenotype expected in the F2 of a dihybrid cross, if 800 seeds are obtained. 3

OR

A concave lens of f = -10 cm forms an image of an object at 20 cm. Find the image distance, nature and magnification. 3

32. Explain why the colour of the Sun is red during sunset but white at noon. 3

OR

Explain with the help of a formula why a thick wire is used for heavy current. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain how the blood sugar level is controlled and what happens in diabetes. 5

34. Explain the structure and functions of the human female reproductive system, and how it supports the development of a baby. 5

35. Explain the terms: genotype, phenotype, dominant, recessive and allele, with the help of the cross Tt x Tt. 5

36. The refractive indices of kerosene, water, glass and diamond are 1.44, 1.33, 1.5 and 2.42. 5
(a) Arrange in increasing order of optical density
(b) find the speed of light in diamond
(c) state in which light bends the most.

OR

A person's far point is at 80 cm and near point at 25 cm. Find the power of the lens needed to see distant objects. Is any lens needed for reading? 5

37. Explain Joule's law of heating. Find the heat produced when a current of 2 A flows through a 50 ohm resistor for 10 minutes. State the heat in calories (1 cal = 4.2 J). 5

OR

A household circuit has a 5 A circuit for lights and fans and a 15 A circuit for power appliances. Describe its layout, state why a geyser of 2 kW needs the 15 A circuit and why a fuse is used. 5

38. Explain the flow of energy in an ecosystem. Why can the energy not be recycled? 5

OR

For 5
(i) $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$ and
(ii) $\mathrm{Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 3\,CO_2}$, identify the substance oxidised, reduced, and the oxidising and reducing agents.

END OF QUESTION PAPER Total: 80 Marks

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